

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology & Geophysics

2. Subject Name: Mineralogical Thermodynamics L-T-P: 3-0-0 Credits:3

3. Pre-requisites: (i) Introduction to Earth Science, (ii) Mineralogy

4. Syllabus and reference books:

Syllabus:

Thermodynamic properties of pure phases: thermodynamic coordinates of a system, state variable, Interrelationships of thermodynamic parameters and functions (volume, heat capacity, enthalpy, entropy, free energy, compressibility and thermal expansivity); Energetics of crystalline solutions: standard states, activity, excess free energy of mixing; Mineralogical phase rule and variance, Graphical representation of mineral assemblages in simple and complex systems; Principles of Schreinemaker's analysis and construction of petrogenetic grids; Heterogeneous systems and reaction equilibria involving solids, geothermobarometry; Thermodynamic properties of fluids: PVTX relations in real gases - Redlich-Kwong and Virial equations of state; mixed volatile equilibria; Thermodynamics of melt-crystal equilibria and trace element modeling; Theory of aqueous electrolytes with applications to study of water-rock interaction.

Reference Books:

- 1) Elementary Thermodynamics for geologists (Wood B and Fraser DG)
- 2) Equilibrium Thermodynamics in Petrology (Powell, R)
- 3) Geochemical Thermodynamics (Nordstorm DK and Munoz JL)
- 4) Applied Mineralogical Thermodynamics (Chatterjee ND)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Thermodynamic properties of pure phases	6
2.	Energetics of crystalline solutions	4
3.	Schreinemaker's analysis	5
4.	Heterogeneous systems and reaction equilibria involving solids	6
5.	Thermodynamic properties of fluids	5
6.	Thermodynamics of melt-crystal equilibria	5
7.	Theory of aqueous electrolytes	5
Total number of hours		36